

## **Minesweeper Oct 02 Testing Document**

EECS 581 Fall 2026

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## Overview

Software Name: Implemented Minesweeper

Tester: Warren Tan

Test Date: 02 Oct, 2026

## Test Cases

| Test ID | Description                    | Method                                                                        | Expected Result          | Actual Result            | Status |
|---------|--------------------------------|-------------------------------------------------------------------------------|--------------------------|--------------------------|--------|
| TC-01   | 0 Mines                        | Set the initial mines to 0                                                    | reject input             | reject input             | Pass   |
| TC-02   | 10 Mines                       | Set the initial mines to 10                                                   | 10 mines are placed      | 10 mines are placed      | Pass   |
| TC-03   | 10.4 Mines                     | Set the initial mines to 10.4                                                 | reject input             | reject input             | Pass   |
| TC-04   | 10.5 Mines                     | Set the initial mines to 10.5                                                 | reject input             | reject input             | Pass   |
| TC-05   | 20 Mines                       | Set the initial mines to 20                                                   | 20 mines are placed      | 20 mines are placed      | Pass   |
| TC-06   | 99 Mines                       | Set the initial mines to 99                                                   | reject input             | reject input             | Pass   |
| TC-07   | First-Click Protection         | Set the initial mines to 20.<br>Start the game. Click a cell.                 | first cell is not a mine | first cell is not a mine | Pass   |
| TC-08   | First-Click Protection w/ Flag | Set the initial mines to 20.<br>Start the game. Flag a cell.<br>Click a cell. | first cell is not a mine | first cell is not a mine | Pass   |

| Test ID | Description | Method                                     | Expected Result                                           | Actual Result                                             | Status |
|---------|-------------|--------------------------------------------|-----------------------------------------------------------|-----------------------------------------------------------|--------|
| TC-09   | Board Size1 | Run buildGrid(10, 10, bombs)               | 10x10 grid created                                        | 10x10 grid created                                        | Pass   |
| TC-10   | Board Size2 | Run buildGrid(10, 11, bombs)               | 10x11 grid created                                        | 10x11 grid created                                        | Pass   |
| TC-11   | Board Size3 | Run buildGrid(11, 10, bombs)               | 11x10 grid created                                        | 11x10 grid created                                        | Pass   |
| TC-12   | Board Size4 | Run buildGrid(11, 11, bombs)               | 11x11 grid created                                        | 11x11 grid created                                        | Pass   |
| TC-13   | Board Size5 | Run buildGrid(9, 9, bombs)                 | 9x9 grid created                                          | 9x9 grid created                                          | Pass   |
| TC-14   | User Wins   | Finish the game with 10 bombs              | User is directed to win screen                            | User is directed to win screen                            | Pass   |
| TC-15   | Lose Music  | Lose a game, then restart to title screen. | Game music is paused on loss, then active on title screen | Game music is paused on loss, then active on title screen | Pass   |

| Test ID | Description      | Method                                                                                                                   | Expected Result                                                   | Actual Result                                       | Status      |
|---------|------------------|--------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|-----------------------------------------------------|-------------|
| TC-16   | Easy AI Solver   | Set the initial mines to 15. Choose "vs AI". Set AI Difficulty to "Easy". Toggle "AI plays automatically". Start game.   | AI chooses tiles at random                                        | AI chooses tiles at random                          | <b>Pass</b> |
| TC-17   | Medium AI Solver | Set the initial mines to 15. Choose "vs AI". Set AI Difficulty to "Medium". Toggle "AI plays automatically". Start game. | AI with smart flagging and opening (detailed in design documents) | AI solved with basic flagging and opening logic     | <b>Pass</b> |
| TC-18   | Hard AI Solver   | Set the initial mines to 15. Choose "vs AI". Set AI Difficulty to "Hard". Toggle "AI plays automatically". Start game.   | Medium AI with 1-2-1 pattern solving.                             | Solved the board while utilising the 1-2-1 pattern. | <b>Pass</b> |

| Test ID | Description                      | Method                                                                                                              | Expected Result                                                                   | Actual Result                                                                   | Status |
|---------|----------------------------------|---------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------|--------|
| TC-19   | Local Multiplayer Bombs          | Set the initial mines to 15. Choose "2 player". Start game.                                                         | Game is started with 15 mines placed.                                             | 15 mines are placed.                                                            | Pass   |
| TC-20   | Local Multiplayer Flags          | Set the initial mines to 15. Choose "2 player". Player 1 flags a tile. Player 2 attempts to remove Player 1's flag. | Flag is placed and cannot be altered by opponent.                                 | Flag is placed. Player 2 cannot remove Player 1's flag.                         | Pass   |
| TC-21   | Local Multiplayer Instant Death  | Set the initial mines to 15. Choose "2 player". Select "Instant Death". Select tiles until mine is selected.        | Game is ended. Opposing player wins.                                              | Game is ended. Player 2 selected the mine and lost.                             | Pass   |
| TC-22   | Local Multiplayer Point Tracking | Set the initial mines to 15. Choose "2 player". Select "Points". Play game as normal while tracking point changes.  | -3 points for mine hit. +1 for flagging a real mine. -2 for flagging a safe tile. | -3 points for mine hit. Flagging points are applied after the board is cleared. | Pass   |
| TC-23   | Local Multiplayer Point Win      | Set the initial mines to 15. Choose "2 player". Select "Points". Play game as normal.                               | The player with the greater points wins.                                          | Player with the greater point count wins game.                                  | Pass   |